

Customer Journey Map

Project Name: Human Development Index (HDI) Prediction Using Machine Learning

Phase	Stage 1 – Data Collection & Input	Stage 2 – Prediction & Analysis	Stage 3 – Decision Making
Actions	• User opens the HDI prediction system. • Enters Life Expectancy, Mean Years of Schooling, Expected Years of Schooling, and GNI Per Capita. • Clicks the Predict button.	• The system preprocesses the data. • The machine learning model predicts the HDI category. • Prediction result is displayed.	• User reviews the predicted HDI category. • Compares results with expected values. • Uses the prediction for planning and analysis.
Touchpoints	• Flask Web Application • Input Form • Dataset	• Machine Learning Model • Prediction Engine • Result Page	• Prediction Dashboard • Reports • Decision Support Interface
Customer Thought	• "I hope the system predicts accurately." • "The interface should be easy to use." • "I want quick results."	• "How is the prediction generated?" • "Is the model accurate?" • "Can I trust these results?"	• "This prediction will help in decision making." • "I can identify development gaps." • "The system saves time."
Customer Feeling	• Curious • Interested • Hopeful	• Confident • Satisfied • Informed	• Happy • Motivated • Confident in planning

Process Ownership

Stage	Owner
Stage 1	User / Data Entry Module
Stage 2	Machine Learning Model & Flask Application

Stage	Owner
Stage 3	User / Government Official / Researcher / Policy Maker

Opportunities for Improvement

Stage	Improvement Opportunities
Stage 1	• Validate user input automatically. • Add input guidelines and tooltips. • Improve the user interface for easier data entry.
Stage 2	• Increase model accuracy with more training data. • Add confidence scores for predictions. • Display feature importance and visualizations.
Stage 3	• Provide country comparison charts. • Generate downloadable reports. • Offer recommendations to improve HDI indicators.

Summary

The customer journey begins when the user enters socio-economic indicators into the application. The system processes the information using a trained machine learning model and predicts the HDI category. Finally, the user reviews the prediction and uses it to support research, policy planning, or educational analysis. This journey highlights a simple, efficient, and data-driven experience for understanding human development.